

SIMI KUMARI

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Summary

AI/ML enthusiast and AI Backend Engineer Intern experienced in building LLM-powered applications, RAG systems, and scalable backend APIs using Python, FastAPI, LangChain, and modern AI frameworks.

Education

IKG Punjab Technical University

Punjab, India

B.Tech in Computer Science and Engineering

Jul 2023 – Jun 2027

CGPA: 8.56/10

Experience

FlyRank AI

Jul 2026 – Present

AI Backend Engineer Intern

Remote

- Developing AI backend services using **Python** and **FastAPI** for AI-powered applications.
- Building and integrating **REST APIs** to support AI workflows and LLM-powered application features.
- Collaborating on scalable backend architectures, database integrations, and API-driven services.
- Contributing to backend development, debugging, and integration of modern AI development tools.

Projects

MedIntel AI – Healthcare Intelligence Platform

[GitHub](#) | [Live Demo](#)

React, TypeScript, FastAPI, PostgreSQL, LangChain, Groq API, Tesseract OCR, Docker

- Developed a full-stack healthcare intelligence platform supporting symptom assessment, medical report interpretation, medication guidance, drug interaction analysis, BMI calculation, and medical image analysis.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline using LangChain, vector embeddings, and LLMs for context-aware assistance from uploaded medical documents.
- Implemented an OCR workflow using **Tesseract OCR** and PyMuPDF for automated medical report extraction and analysis.
- Built authentication, persistent conversation history, and backend services using **FastAPI** and **PostgreSQL**.

AI Learning Companion – RAG Conversational AI

[GitHub](#) | [Live Demo](#)

LangChain, ChromaDB, SentenceTransformers, Groq API, Streamlit

- Engineered an end-to-end **RAG pipeline** combining semantic search, vector embeddings, retrieval, and LLM-based response generation.
- Developed document chunking and embedding workflows for processing and indexing **100+ PDF knowledge sources**.
- Implemented semantic vector search using **ChromaDB**, achieving retrieval latency under **2 seconds**.

AI News Classifier – Transformer-based NLP Application

[GitHub](#) | [Live Demo](#)

Python, PyTorch, BERT, Hugging Face Transformers, Streamlit

- Fine-tuned a **BERT** transformer model for automated news classification, achieving **92% accuracy**.
- Processed and classified **10,000+ text samples** using preprocessing and tokenization pipelines.
- Developed an inference pipeline capable of generating predictions in **under one second**.

Technical Skills

Languages: Python, SQL

AI/ML: Machine Learning, Deep Learning, NLP, Transformers, BERT, LLMs, Generative AI, RAG, Prompt Engineering

GenAI & Retrieval: LangChain, ChromaDB, Vector Databases, Vector Embeddings, Semantic Search, SentenceTransformers

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Pandas, NumPy

Backend & APIs: FastAPI, REST APIs, Groq API, Hugging Face API

Databases: PostgreSQL, ChromaDB

Tools & Deployment: Git, GitHub, Docker, Streamlit, Render, Vercel Hugging Face Spaces

Soft Skills: Problem Solving, Analytical Thinking, Communication, Team Collaboration, Adaptability

Certifications

- Data Science with Python – Internshala
- Complete Data Science with Machine Learning – Independent learning
- Complete Generative AI – Udemy